What Makes a Movie Successful?

Jin Zhang | CSE 163 | Exploratory Data Analysis

GitHub repository: Not created yet — add the public repository URL before submission.

Summary of Research Questions

1. Is movie budget associated with box office revenue, and does the relationship vary by genre? 2. Is the number of listed cast members associated with revenue? 3. Are frequently occurring movie keywords associated with revenue? All questions concern associations, not causal effects.

Motivation

Movies require substantial financial investment, and simple features such as budget, cast size, genre, and themes may be related to commercial outcomes. Exploring these patterns can describe the film dataset and its limits without implying that any feature causes higher revenue.

Dataset

The Movies Dataset from Kaggle contains movie metadata, credits, and keywords. This EDA uses the provided movies_metadata.csv, credits.csv, and keywords.csv. Metadata includes IDs, budget, revenue, genres, and release details; credits includes cast and crew; keywords lists movie themes. Rows in the first two are intended as movie records; duplicate and unmatched IDs require care. The analysis joins metadata separately to credits and keywords by movie ID.

Method

A runnable Python script reads the original CSV files without changing them, parses list-like genre, cast, and keyword fields, validates numeric IDs, converts budget and revenue, and retains positive values for financial comparisons. A movie may have multiple genres or keywords, so it appears in each applicable category. Spearman rank correlation (SciPy) summarizes monotonic associations. Extreme positive values are retained and distributions are shown on logarithmic axes where useful. The generated plots and summary text are in eda_output/.

EDA Results

Dataset size and join checks

movies_metadata.csv: 45,466 rows × 24 columns. credits.csv: 45,476 × 3. keywords.csv: 46,419 × 2. Metadata has 45433 distinct numeric IDs after invalid IDs are excluded and 30 duplicate valid IDs encountered. Joining metadata to credits produced 45,476 credit rows with 45,432 distinct matching IDs; joining metadata to keywords produced 46,419 rows with 45,432 distinct matching IDs. Joined row counts can exceed distinct IDs because source files contain repeated movie IDs.

Metadata selected fields blank: budget: 0 (0.0%); genres: 0 (0.0%); id: 0 (0.0%); revenue: 6 (0.0%)

Credits selected fields blank: cast: 0 (0.0%); id: 0 (0.0%)

Keywords selected fields blank: id: 0 (0.0%); keywords: 0 (0.0%)

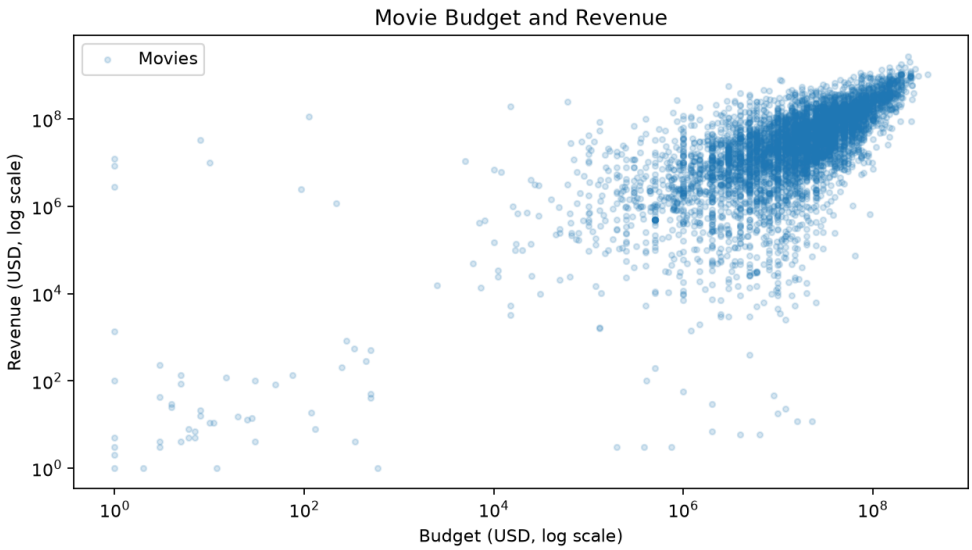
Blank cells are counted using empty-string checks. Structured text that is malformed or represents an empty list is not captured by this blank-only measure; parsing failures are treated as empty lists by the script. For financial analysis, only positive numeric budget and revenue are used, leaving 5,375 movies. This filters missing, zero, negative, and nonnumeric values; it does not remove large positive outliers.

Variables and summaries

Budget and revenue (metadata): among movies with both positive values, budget — n=5,375, mean=31,108,015.75, SD=40,176,525.28, min=1.00, Q1=5,053,316.00, median=17,000,000.00, Q3=40,000,000.00, max=380,000,000.00; revenue — n=5,375, mean=90,391,821.65, SD=166,202,762.00, min=1.00, Q1=7,020,199.00, median=29,918,745.00, Q3=100,000,000.00, max=2,787,965,087.00. **Cast size (credits joined to metadata):** number of parsed cast entries per row — n=45,476, mean=12.37, SD=12.09, min=0.00, Q1=6.00, median=10.00, Q3=15.00, max=313.00. Genre is categorical and multi-valued; among financially usable movies the most frequent categories are Drama (2,583), Comedy (1,851), Thriller (1,502), Action (1,414), Romance (1,013), Adventure (957), Crime (861), Science Fiction (634), Horror (586), Family (530).

Budget–revenue Spearman $\rho = 0.710$ ($p < .001$); cast-size–revenue $\rho = 0.419$ ($p < .001$). SciPy returned values smaller than its displayed floating-point range, so p-values are reported as < .001. These results support positive monotonic associations in the usable rows, not causation. The keyword comparison focuses on frequent terms; the output includes each selected term’s group size and revenue summary. Groups overlap and are not adjusted for budget or genre.

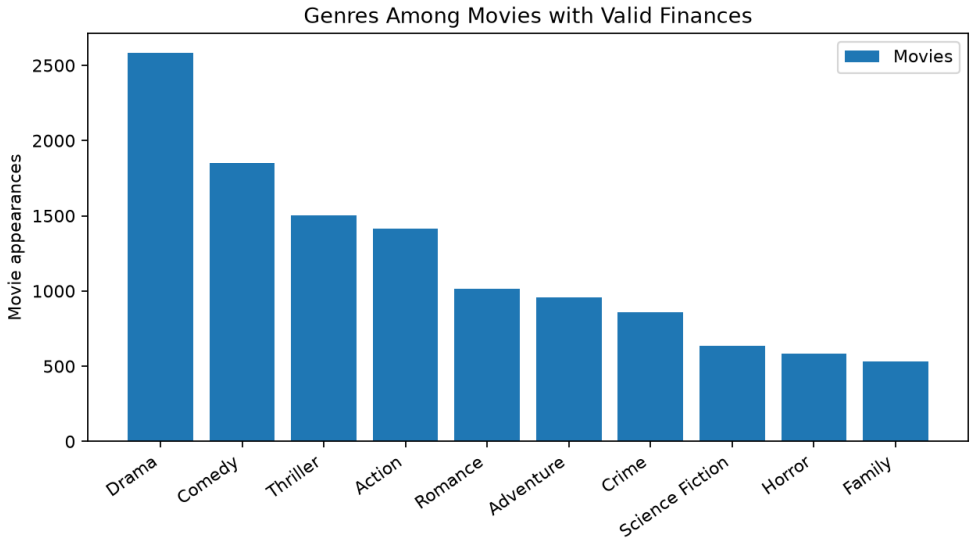
Figure 1



Caption: Usable positive values appear on log scales; points show association, not causation.

Figure 1. Budget and revenue in the metadata subset. The log-log scatter plot preserves the wide range of positive values; the visible upward pattern is consistent with the positive rank correlation.

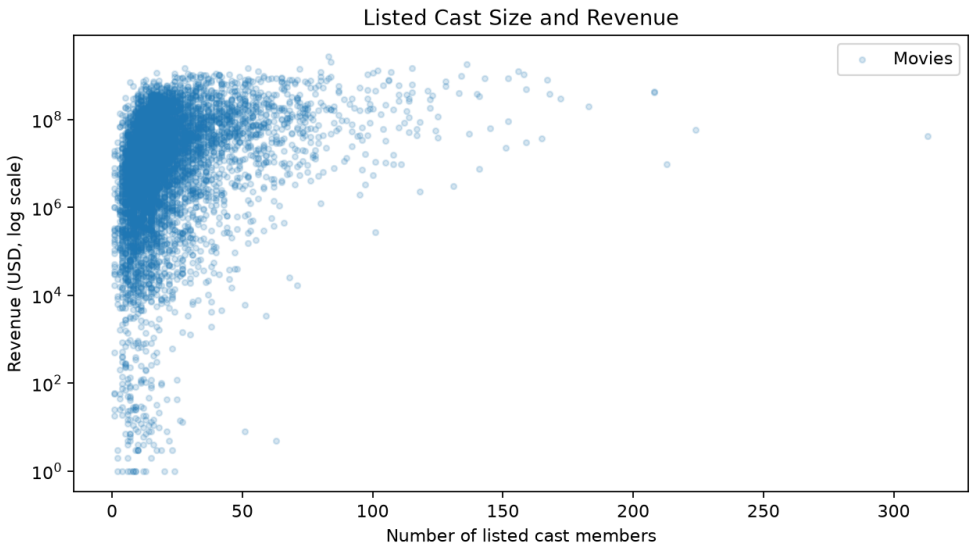
Figure 2



Caption: Genre counts can exceed the number of movies because each movie may have multiple genres.

Figure 2. Genre counts among movies with valid positive budget and revenue. Multi-genre movies contribute to each listed genre, so counts are appearances rather than exclusive groups.

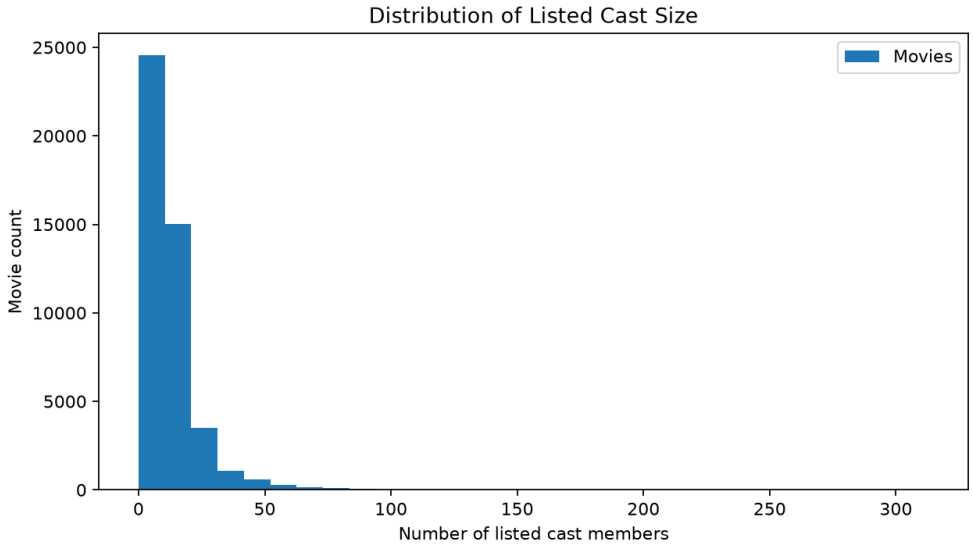
Figure 3



Caption: Revenue varies widely at each cast size; only positive revenue and nonempty parsed casts are included.

Figure 3. Cast size and revenue after joining credits to metadata. The wide vertical spread indicates that cast size alone does not determine revenue.

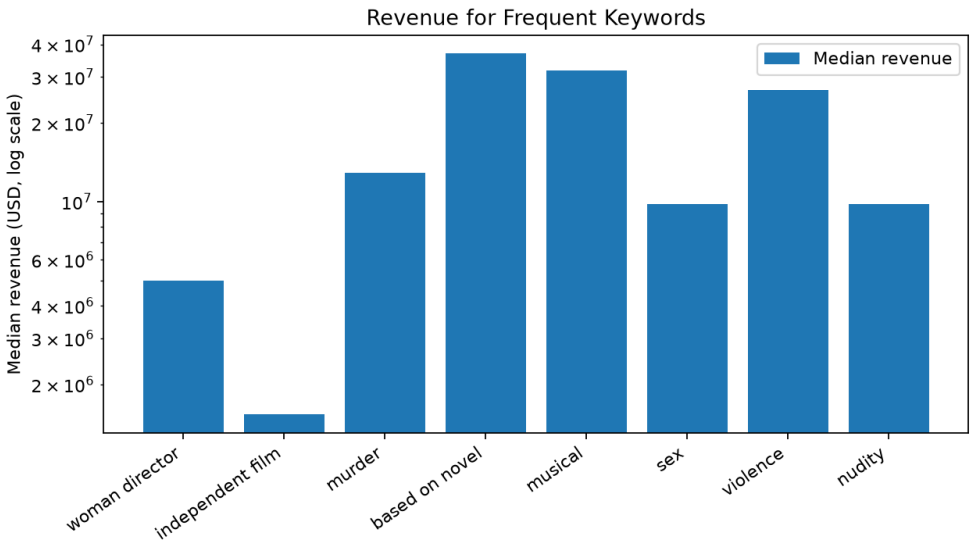
Figure 4



Caption: The distribution describes cast entries in the credits file and does not establish that every cast list is complete.

Figure 4. Distribution of parsed cast-list lengths. Counts reflect names listed in the file and may not represent complete casts.

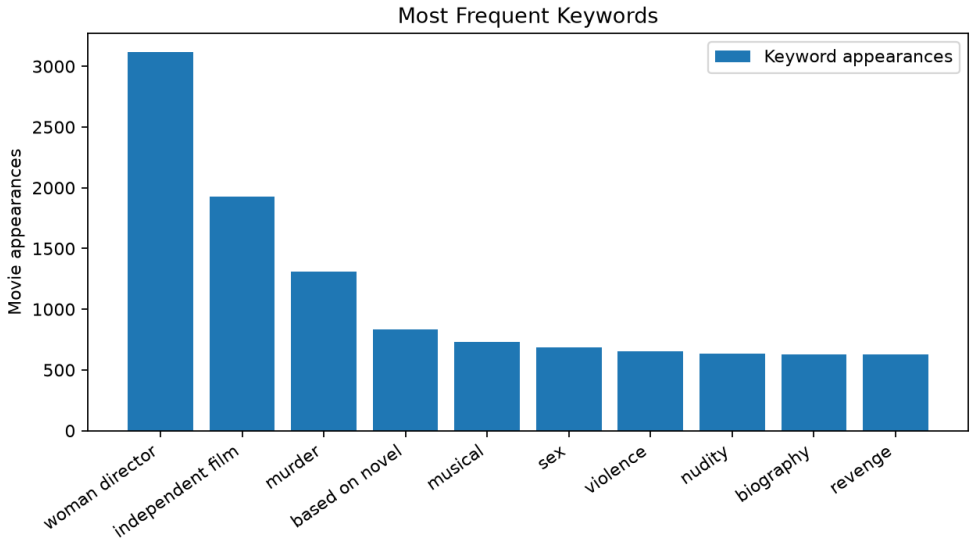
Figure 5



Caption: Median revenue for keywords in at least 100 joined rows; groups overlap because movies have multiple terms.

Figure 5. Median revenue for frequent keyword groups. These overlapping, unadjusted groups are descriptive; differences may reflect budget, genre, release era, or other factors.

Figure 6



Caption: Keyword names were lowercased and spaces normalized before counting.

Figure 6. Ten most frequent normalized keyword terms in the joined keyword data. This plot describes coverage and helps explain which terms are eligible for comparison.

Challenge Goals Update

Multiple datasets: metadata is joined separately with credits and keywords. Both combinations are explored with two plots each; ID matching and source row counts are reported above. This satisfies the intended use of three source datasets across two joins, though it does not create one three-way table. **New library (SciPy):** scipy.stats.spearmanr is used for budget/revenue and cast-size/revenue. The EDA treats significance cautiously and does not claim causal evidence.

Work Plan Evaluation

The proposal estimated 17 hours across inspection, cleaning, joining, analysis, visualization, testing, and writing. A reliable time log was not available, so actual hours cannot be compared honestly. Completed here: file inspection, data cleaning and joins, descriptive summaries, six plots, two correlation analyses, helper-function tests, and this report draft. Remaining: add the public GitHub URL, verify course staff feedback and dataset-use restrictions, and review the final PDF. Estimate 1–2 hours.

Testing

test_eda.py uses assert statements on small constructed examples to check list parsing, numeric cleaning, invalid IDs, ID-based joins, keyword normalization, and numeric summaries. These tests isolate edge cases from the full dataset. The full-data run produced the counts, plots, and correlations summarized above. Plot generation also provides a visual inspection step; results should be rerun from the unchanged source CSVs to reproduce them.

Collaboration

I consulted the CSE 163 Part 2 EDA assignment instructions, my Part 1 proposal, Python’s standard library, Matplotlib, and SciPy documentation. No other people were consulted for this EDA draft.

Limitations and next steps

The source files contain duplicates, missing values, malformed structured text, and potentially inconsistent IDs. Revenue and budget are highly skewed and may be recorded in different historical contexts. Cast-list size is not the same as production cast size. Keyword and genre groups overlap; keyword medians are not adjusted for confounding. Next analysis should inspect genre-specific budget–revenue relationships, compare revenue across cast-size bands, and assess keyword groups with appropriate controls and sufficient sample sizes.